

Team Hawkeye at Oleekbochon: A Hybrid Program-Aided Reasoning and RAG-Augmented Fine-Tuning Framework for Bengali LLM Hallucination Detection*

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Abstract

Large Language Models frequently generate fluent yet factually incorrect statements—a problem especially acute in low-resource languages such as Bengali. We present Team Hawkeye’s solution for the *Oleekbochon Bengali LLM Hallucination Detection Challenge*. Our framework integrates: (1) a Self-Consistency Thought-augmented Reasoning (SC-TIR) code execution engine with Bengali numeric parsing for mathematical queries; (2) a dense-sparse RAG pipeline using BGE-M3 bi-encoders and cross-encoder rerankers over a 250k+ chunk Bengali Wikipedia corpus; (3) probabilistic synthetic data generation (100k samples) across four hallucination categories; and (4) a fine-tuned Qwen3.5-4B logprob classifier via vLLM. Our system achieved **0.83834 Macro F1** (13th place), improving **+16 ranks** over Phase 1.

Keywords: Hallucination Detection, Bengali NLP, RAG, Program-Aided Language Models, SFT, Oleekbochon.

1 Introduction

Large Language Models have achieved remarkable NLP advancements, yet their tendency to generate plausible but factually inaccurate assertions—*hallucination*—remains a fundamental deployment obstacle in high-stakes domains such as education, law, and healthcare (Ji et al., 2023). While extensive research has focused on hallucination detection in English, low-resource and morphologically complex languages face severe vulnerabilities.

Bengali, spoken by over 270 million people worldwide, presents unique computational challenges for LLM factual verification, notably complex numeric representations leading to arithmetic matching failures, culturally situated knowledge

deficits in generic models, and high idiomatic ambiguity.

The **Oleekbochon Challenge** (Islamic University of Technology Computer Society (IUTCS), 2026), hosted by IUTCS, tasks participants with binary faithfulness prediction for Bengali question-response pairs. Our contributions are: **(1) Program-Aided Math Reasoning:** A deterministic math query router and Python executor; **(2) Dense-Sparse RAG:** A retrieval engine over 250k+ Wikipedia chunks using BGE-M3 and BGE-Reranker-v2; **(3) Probabilistic Synthetic SFT:** 100,000 training pairs with K -chunk distractor context mixing; and **(4) Fine-Tuned Verification:** Qwen3.5-4B logprob classification.

Our framework propelled Team Hawkeye from 29th to **13th place** (+16 ranks) with **0.83834 Macro F1**.

2 Related Work

Current hallucination detection methods typically rely on self-consistency metrics, NLI entailment scoring, or reference-based retrieval (Manakul et al., 2023; Es et al., 2023). Since standard cross-encoders often struggle with morphologically rich languages, Retrieval-Augmented Generation (RAG) is used to enhance factual grounding by injecting relevant contexts (Lewis et al., 2020). Models like BGE-M3 (Chen et al., 2024) support effective multilingual retrieval, and combining bi-encoders with cross-encoder reranking significantly improves precision. Furthermore, for reasoning tasks, Program-Aided Language Models (PAL) (Gao et al., 2023) decouple logic synthesis from arithmetic evaluation by offloading computation to code interpreters, an approach we adapt for Bengali using custom heuristics and numeric converters.

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Table 1: Empirical Pipeline Evaluation across LLM Backbones and Verification Paradigms

Pipeline Architecture	Backbone LLM	Training & Feature Configuration	Public F1	Private F1
Direct SFT Classifier (Best)	Qwen3.5-4B	5k Augmented + RAG + SC-TIR Math	0.901	0.903
Direct SFT Classifier (9B Scale)	Qwen3.5-9B	10k Augmented + RAG + SC-TIR Math	0.899	0.894
Untrained Few-Shot Baseline	Gemma-4-26B	Few-Shot + RAG + SC-TIR Math	0.894	0.883
SFT Hunter + Fine-Tuned NLI Judge	Qwen3.5-4B	5k Augmented + Fine-Tuned NLI Judge	0.868	0.876
SFT Hunter + Pretrained NLI	Qwen3.5-4B	5k Augmented + Pretrained NLI Classifier	0.869	0.859
Few-Shot Baseline (No RAG/Math)	Gemma-31B	Few-Shot (Without RAG & Math Solver)	0.834	0.843
SFT on Raw Synthetic (100k)	Qwen3.5-9B	100k Un-augmented SFT (qwen_train) + RAG	0.800	0.795
One-Shot Baseline (No RAG/Math)	Gemma-31B	1-Shot (Without RAG & Math Solver)	0.694	0.683

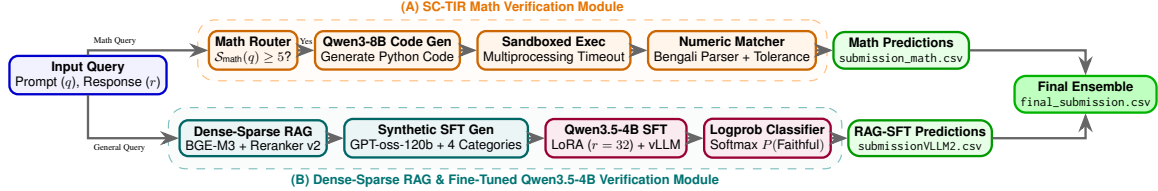


Figure 1: Overview of Team Hawkeye’s Multi-Stage Hallucination Verification Framework.

3 System Architecture

Our system (Fig. 1) operates as a multi-stage pipeline: queries are first routed through the SC-TIR Math module; remaining queries are enriched via the RAG engine, processed by the SFT classifier, and ensembled into the final submission.

3.1 Stage 1: SC-TIR Math Verification Engine

Bengali math word problems (e.g., age ratios, interest rates) are highly susceptible to tokenization errors. We design a dedicated Program-Aided Reasoning pipeline.

3.1.1 Math Query Routing

A scoring function $\mathcal{S}_{\text{math}}(q)$ is computed using Bengali digit occurrences, arithmetic operators, ratio patterns, and domain keywords ($\mathcal{K}_{\text{math}}$: *onupat, gawd, shotkora, labh, khoti, taka*):

$$\begin{aligned} \mathcal{S}_{\text{math}}(q) = & w_d \cdot \mathbb{I}(q \text{ has digits}) \\ & + \sum_{k \in \mathcal{K}_{\text{math}}} w_k \cdot \mathbb{I}(k \in q) \\ & - w_h \cdot \mathbb{I}(q \text{ has date terms}) \end{aligned} \quad (1)$$

Queries with $\mathcal{S}_{\text{math}}(q) \geq 5$ are routed to the code executor.

3.1.2 Code Generation and Execution

Routed queries are sent to Qwen3-8B via vLLM to generate Python code saving the result to `final_answer`. Code executes inside an isolated multiprocessing worker with a 5-second timeout.

3.1.3 Bengali Numeric Matching

Answers are clustered via tolerance matching:

$$|v_1 - v_2| \leq \max(\epsilon_{\text{abs}}, \epsilon_{\text{rel}} \cdot \max(|v_1|, |v_2|)) \quad (2)$$

where $\epsilon_{\text{rel}} = 10^{-3}$, $\epsilon_{\text{abs}} = 10^{-2}$. A deterministic Bengali number word parser converts units (0–99) and scale multipliers (*shot*=100, *hajar*=1000, *lakh*= 10^5 , *koti*= 10^7) to numeric values. If the computed answer $\hat{y}$ matches any candidate in the response, the label is Faithful; otherwise Hallucinated. Algorithm 1 formalizes the pipeline.

3.2 Stage 2: Dense-Sparse RAG Engine

Non-mathematical queries require external evidence for factual verification.

3.2.1 Knowledge Base Construction

We construct a Bengali knowledge base aggregating 12 multi-domain datasets: TituLM Bangla MMLU (Nahin et al., 2025), Bengali Wikipedia, Bangla Bagdhara (Adib et al., 2026b), BenHalluEval (Adib et al., 2026a), BLAD (Sakhawat, 2025), NCTB textbooks (Class 8–10), Bangladesh Open University books, BUCK benchmark (Kabir et al., 2025), Government Bangladesh Lexicon (Government of Bangladesh, 2024), BanglaVerb (Ayman et al., 2025b), BanglaBlend (Ayman et al., 2025a), and BanglaTense (Bijoy et al., 2024). Raw texts are NFKC-normalized and chunked into passages of 500 characters with 100-character overlap using sentence-aware boundary detection, yielding 250,000+ indexed chunks.

Algorithm 1 SC-TIR Math Verification Pipeline

Require: Query q , Response r , Model $\mathcal{M}_{\text{code}}$, Tolerance ϵ

Ensure: Binary Label $y \in \{0, 1\}$ or Fallback

```
1:  $s \leftarrow \mathcal{S}_{\text{math}}(q)$ 
2: if  $s < 5$  then
3:   return FALLBACK_TO_RAG
4: end if
5:  $\text{Code} \leftarrow \mathcal{M}_{\text{code}}.\text{Generate}(\text{BuildSCPrompt}(q))$ 

6:  $\hat{y} \leftarrow \text{SandboxedExecute}(\text{Code}, 5s)$ 
7: if  $\hat{y}$  is NULL then
8:   return FALLBACK_TO_LLM_JUDGE
9: end if
10:  $\mathcal{C} \leftarrow \text{ExtractNumericCandidates}(r)$ 
11: if  $\text{ValuesMatch}(\hat{y}, \mathcal{C}, \epsilon)$  then
12:   return  $y = 1$ 
13: else
14:   return  $y = 0$ 
15: end if
```

3.2.2 Dense Indexing and Reranking

Passages are embedded into 1024-dim vectors using BAAI/bge-m3 (FP16, batch size 2048) and L_2 -normalized:

$$\mathbf{v}_p = \frac{\mathbf{e}(p)}{\max(\|\mathbf{e}(p)\|_2, 10^{-12})} \quad (3)$$

Embeddings are indexed into FAISS Flat Inner Product (IndexFlatIP) in blocks of 100k vectors, ensuring 100% recall. For each query q , the top $M=40$ candidate passages are retrieved:

$$\mathcal{P}_{\text{topM}} = \arg \max_{p \in \text{KB}}^{(M)} \left(\mathbf{v}_q^\top \mathbf{v}_p \right) \quad (4)$$

The query-passage pairs (q, p_i) are then reranked using BAAI/bge-reranker-v2-m3; the top $K=5$ reranked passages are extracted, formatted with reference indices, and concatenated into the retrieved context.

3.3 Stage 3: Synthetic Data and SFT Augmentation

To fine-tune a robust classifier without relying solely on limited ground-truth data, we create a large synthetic dataset using openai/gpt-oss-120b via the Tinker API.

3.3.1 Hallucination Category Generation

Given factual Bengali context snippets, the teacher model generates four error categories:

1. **Entity Swap:** Swapping names, locations, or organizations with related terms.
2. **Numeric Alteration:** Modifying dates, quantities, or counts.
3. **Contradiction:** Inverting the logical truth of a factual claim.
4. **Baseless Addition:** Introducing realistic yet unmentioned details.

3.3.2 Probabilistic Context-Mixing

To prevent positional bias, we apply four mixing operations: baseline clean context (30%), single distractor swap with fallback token [Information Not Found] (15%), target mixed with $K-1$ distractors in shuffled order (40%), and distractors-only without target (15%). From 100,000 augmented samples, 5,000 balanced samples are selected for fine-tuning.

3.4 Stage 4: Verification Model and Inference

3.4.1 SFT Information Hunter

We train Qwen3.5-4B via LoRA ($r=32$, $\eta=2 \times 10^{-4}$) as an *Information Hunter*: given question q and RAG context, the model extracts supporting evidence verbatim or returns the fallback token [Information Not Found] (*tottho paowa jayni*).

3.4.2 NLI Verification Backends

Once evidence e is extracted, we evaluate two NLI paradigms: (1) a fine-tuned cross-encoder judge accepting (Premise= e , Hypothesis= r) for binary entailment, and (2) an off-the-shelf multilingual NLI classifier as baseline.

3.4.3 Logprob Classification and Inference

For direct verification, candidate logprobs $\mathcal{L}_{\text{true}}$ and $\mathcal{L}_{\text{false}}$ are Softmax-normalized:

$$P(\text{Faithful}) = \frac{\exp(\mathcal{L}_{\text{true}})}{\exp(\mathcal{L}_{\text{true}}) + \exp(\mathcal{L}_{\text{false}})} \quad (5)$$

Prompts are dynamically truncated via `fit_retrieved_context_to_budget` for Tesla T4 VRAM limits. Final predictions combine SC-TIR math and SFT verification outputs.

4 Experimental Setup and Results

4.1 Dataset and Evaluation Metric

The competition test dataset consists of 2,598 test rows containing Bengali questions, candidate re-

Table 2: Official Phase 2 Final Standings (Oleekbochon Challenge)

Rank	Team Name	Macro F1-Score
1	BUET_NeuralX	0.95424
2	Team_ReActive	0.92392
3	DU_RDanto	0.92120
...	...	...
13	Hawkeye (Ours)	0.83834

sponses, and optional context snippets. Performance is evaluated strictly using Macro F1-score:

$$\text{Macro F1} = \frac{\text{F1}_{\text{Faithful}} + \text{F1}_{\text{Hallucinated}}}{2} \quad (6)$$

4.2 Official Competition Standings

Table 2 presents the official Phase 2 standings. Team Hawkeye achieved a final Macro F1-score of **0.83834**, securing **13th place** out of 32 qualified teams.

4.3 Empirical Pipeline Comparison and Comparative Analysis

To systematically evaluate our architectural design choices, we conduct extensive empirical experiments across multiple LLM backbones (Qwen3.5-4B, Qwen3.5-9B, Gemma-4-26B, Gemma-4-31B) and verification paradigms (Direct SFT Logprob Classification vs. Fine-Tuned NLI Judge vs. Pretrained NLI Classifier vs. Un-augmented SFT). Table 1 summarizes the Public and Private test set performance across all evaluated system configurations.

1) Superiority of Direct Logprob Classification: Our fine-tuned Qwen3.5-4B model trained on 5,000 synthetic samples with probabilistic distractor mixing achieved the highest overall score of **0.901 Public F1** and **0.903 Private F1** when evaluated directly via logprob Softmax normalization.

2) Impact of Probabilistic Distractor Mixing (+0.108 F1 Jump): A crucial finding lies in comparing direct fine-tuning on raw synthetic data versus probabilistic distractor mixing. Training Qwen3.5-9B directly on 100,000 raw un-augmented synthetic pairs generated by `generate_synthetic_data.ipynb` via `qwen_train.ipynb` achieved only **0.800 Public / 0.795 Private F1**. Without distractor mixing, the model overfit to single-passage context layouts and learned keyword shortcuts. Introducing probabilistic K -chunk distractor context mixing (AUGEMENT synthetic sft dataset.ipynb) propelled accuracy by **+0.108 F1 points** to **0.903 Private F1**,

proving that context mixing is essential for multi-reference factual verification.

3) The “Information Hunter” Task Misalignment Paradox: A particularly notable phenomenon emerged regarding the *Information Hunter* paradigm. The Qwen3.5-4B model was specifically fine-tuned to extract supporting evidence verbatim or return [Information Not Found] (*tottho paowa jayni*). However, when its extracted evidence was passed downstream to a dedicated fine-tuned NLI judge (`train_nli_judge.ipynb`), the pipeline achieved **0.876 Private F1**—performing *worse* than using the Information Hunter LLM directly as a direct logprob classifier (**0.903 Private F1**). The model developed superior internal representations for direct factual classification while learning evidence extraction than when used for the explicit downstream extraction task.

4) Ablation of RAG Retrieval and SC-TIR Math Solver: Omitting both the Dense-Sparse RAG engine and SC-TIR Math Solver caused severe performance degradation. An un-augmented few-shot model scored **0.843 Private F1**, while a one-shot model dropped to **0.683 Private F1**, demonstrating that external factual grounding and deterministic code execution are indispensable for Bengali LLM hallucination detection.

5 Conclusion and Future Work

We presented Team Hawkeye’s 13th-place solution for the *Oleekbochon Bengali LLM Hallucination Detection Challenge*. Our framework integrates Program-Aided Language reasoning, dense-sparse BGE-M3 retrieval, probabilistic synthetic SFT data generation, and fine-tuned Qwen3.5-4B logprob classification, yielding a +16 rank progression. A key empirical insight is that direct logprob classification outperforms decoupled intermediate evidence extraction pipelines. Future work will explore scaling to larger backbone models (e.g., Gemma-4 31B), incorporating chain-of-thought self-verification, and extending the framework to other low-resource South Asian languages.

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